CPRE 4310

Homework #5

Assignments will be submitted in PDF format via Canvas.

Please submit your homework online through Canvas. Late homework will not be accepted.
Important: Your submission must be in .pdf format ONLY!
You are not allowed to use AI tools to answer any of the HW questions

1. Describe the differences between a host-based IDS and a network-based IDS. How can their advantages be combined into a single system?

- Host-based IDS
  - A host-based IDS is an agent or software that is installed on a single host.
    - Analyses system logs
    - Checks for any unauthorized changes
    - Monitors system calls
- Network-based IDS
  - A network-based IDS monitors the traffic on the network and identifies suspicious activity occuring.
  - Can be implemented Inline or Passively

Their advantages can be combined into a single system by taking both the alerts generated by Host-based IDS and Network-based IDS, then feeding them into a Security Information and Event Mangement (SIEM) based system. Using this the security admins or the people analyzing the logs will be able to get a much more accurate view of the security of the system as a whole.

2. What are three benefits that can be provided by an IDS?

- Detects malicious activity early
  - Does this by monitoring host or network traffic to identify intrusions or attacks
- Collects logs and other forensics data
  - Captures packet data which can help with forensics later on to dive deeper into identifying the attack vector
- Real-time alerting and allowing quicker response from security administrators
  - Provides alerts the moment any suspicious activity is detected

3. **What is the difference between a false positive and a false negative in the context of an IDS?**

   False Positive is when an alert is created even though the activity is from a genuine user. False Negative is when an alert is not created even though the activity is from an attacker. In other terms, false positive (false alarm) occurs when our IDS wrongly flags legit activity of a user and a false negative occurs when our IDS fails to detect a threat.

4. **What is the difference between anomaly detection and signature intrusion detection?**

   - Anomaly detection
     - This method of detection identifies threats by comparing them against the actions of a normal user.
     - It basically establishes a baseline of what normal user and network behavior looks like, then monitors the activity to see if anything deviates from the standard baseline.
   - Signature intrusion detection
     - This method of detection identifies threats by comparing them against a database of known attack "signatures".
     - The "signatures" used are simply unique patterns or blueprints of malicious code and network traffic stored in a database, which is then used and compared against to see if any of the incoming data matches contents from the database.